Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/26/2014
PI Name:	Stolc, Viktor Ph.D.		
Project Title:	Electromagnetic Basis of Metabolism and Heredity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
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COI Name (Institution):	Freund, Friedemann (SETI Institute)		
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Task Description:	We propose to investigate how the Earth's natural electromagnetic field in the ultralow to extremely low frequency (ULF/ELF) range interacts with crucial reaction steps in biochemical processes that have time constants in the ULF/ELF frequency range. We propose to address the temporal coordination of fundamental biochemical reactions in living cells, which are maintained by the sequential transfer of electrons in reduction-oxidation reactions. This redox cycle (also called the metabolic cycle) is known to control all cellular responses to environmental factors, including those caused by spaceflight-induced stress. In fact, the redox cycle is the fundamental biochemical process that controls the timing of all biochemical reactions in living cells, including energy production, RNA transcription, and DNA replication. Through this temporal coordination chemical conflict between the reductive and oxidative reactions is avoided. At the same time the coherent or interfering transfer of energy is enabled between the ambient natural electromagnetic (EM) field and living cells. We will study how the EM field controls the timing of the biochemical redox cycle and thereby controls heredity. Our main tool to carry out this work will be an enclosure (chamber) with an electronic circuit enabling in real-time, through a dynamic feedback loop, active attenuation, or cancelation or modulation of the ambient EM field.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of the proposed work to confirm EM resonance as an environmental unit factor that affects, or perhaps even drives metabolism, growth, and genome integrity is that confirmation of this new understanding about the ambient Earth and spaceflight environments would enable the previously unattainable goal of mitigating the negative effects of spaceflight on human physiology beyond the lower Earth orbit (LEO). Therefore, successful execution of this proposal can enable human space exploration beyond LEO by advancing fundamental understanding of metabolism on Earth and in spaceflight.
	2014 FSB grant report: Electromagnetic Basis of Metabolism and Heredity. PI: Viktor Stolc; Co-I: Friedemann Freund Living organisms control their ultradian cellular biological clocks to maintain functional oscillation of the redox cycle, also called the “metabolic cycle” or “respiratory cycle.” Organization of cellular processes requires parallel processing on a synchronized time-base. These clocks coordinate the timing of all biochemical processes in the cell, including energy production, DNA replication, and RNA transcription. The period length of the redox cycle varies with the rate of electron flux through the electron transport chain (ETC) and, correspondingly, with energy synthesized in the form of adenosine triphosphate (ATP). Additionally, the effective concentration of ATP oscillates in time because the electron transport along the ETC occurs in a timed, oscillatory manner through the redox enzyme NADPH oxidase. This universal time keeping function can be perturbed by an exogenous induction of reactive oxygen species (ROS). When this happens the metabolic rates changes. This causes oxidative stress, aging, and mutations. Therefore, smooth temporal coordination of the redox cycle not only prevents chemical conflict between the reductive and oxidative partial reactions; it also maintains genome integrity and lifespan. However, this universal biochemical rhythm can also be disrupted by ROS induction in vivo. This disruption can be achieved by blocking the electron transport chain either endogenously or exogenously by various metabolites, e.g. hydrogen sulfide (H2S), highly diffusible drugs, carbon monoxide (CO), and/or acetaldehyde. One of the most significant discoveries of this work is that the in vivo electron transport can be attenuated via a coherent or interfering transfer of energy from ultralow frequency (ULF) and extremely low frequency (ELF) electromagnetic (EM) fields. Such ULF and ELF fields are omnipresent and have probably been omnipresent since the dawn of Life. Recognizing that such a coupling exists between ambient ULF and ELF point to the possibility that Life has not only adapted to such EM fields but may have learned to use them as a time-setting function for universal biochemical, energy-producing reactions in living cells. The ULF and ELF waves, known as Schumann resonances (SR), are a global phenomenon. They are standing waves in the cavity formed by the Earth surface and the ionospheric reflector. Their amplitudes fall into the pico-Tesla range, about 5-6 orders of magnitude weaker than the Earth dipole field (25-65 micro-Tesla depending on latitude). They are maintained and constantly replenished by lightning strikes around the globe. While the SR normally occur at discrete frequencies, 7.8 Hz for the fundamental, 14.3, 20.8, 27.3, and 33.8 Hz for the overtones, they are affected by the diurnal cycle, by the moon’s gravitational forces, by solar cycles and geomagnetic storms as well as by cosmic events such as X-ray bursts. Some of the SR modes are also affected, at least regionally, by seismic events such as large earthquakes. We are studying how biological systems can harness EM waves, such as the SR, and use them as a “tuning fork” for universal biochemical organization in vivo. We have developed an apparatus enabling us to demonstrate the transfer of information and energy from the ambient ULF and ELF fields to the universal electron transport chain (ETC) in living cells. The transfer of electrons along the ETC controls the reduction-oxidation reactions that are essential for fundamental biochemical processes in all living cells. In order for these processes to work properly, the ETC has to maintain some form of synchronization, or coherence, with all biochemical reactions in the living cells, including energy production, RNA transcription, and DNA replication. As a consequence of synchronization, harmful chemical conflict between the reductive and oxidative partial reactions can be minimized or avoided. At the same time synchronization allows for transfer of energy to occur via coupling to the natural ambient EM field – a process that can be constructively coherent or destructively interfering. Bad timing of the synchronization can have a detrimental effect, via increased oxidative damage, on DNA replication, which controls heredity. We now know that the universal cellular metabolic systems are tied to a metronomic time-base (e.g. for yeast grown in rich media, a 40-min interval) that provides a unifying principle – based on temporal ordering – via feedback loops at many different time scales simultaneously, testifying to the enormous complexity of the living state. Remarkably, the genome-wide activity, i.e. RNA transcriptome, also oscillates with the same frequency as the redox cycle, i.e. in phase with the rates of O2 consumption and H2S production; with periodic genes activated in phase either with the oxidative or with the reductive phase. Respiratory oscillations that alternate between an oxidative phase (high oxygen uptake) and a reductive phase (low oxygen uptake) can be conveniently monitored via residual dissolved oxygen (DO) and/or reactive oxygen species (ROS) measurements. These correlations demonstrate a level of temporal organization of genome-wide RNA transcription activity not previously recognized, induced in each gene exclusively by the reductive vs. oxidative cellular state. Moreover, the redox cycle of individual cells grown in batch culture can be modulated not only in amplitude but also synchronized among initially unsynchronized cells by signals that originate in the external physical environment, particularly when such perturbations are punctuated in phase, or in resonance, with the endogenous redox cycle. For example, external perturbations of the redox cycle can be modulated both in amplitude and periodicity by

Task Progress:	highly diffusible drugs, H2S, CO, acetaldehyde, and – our research – by EM fields. Disturbances of the electron “dwell times” in the ETC are directly proportional to the electron leakage and concurrent oxidative stress through the generation of reactive oxygen species (ROS) such as superoxide ion (O2–) and hydrogen peroxide (H2O2). In view of the potential for a resonance-based perturbation of the ETC electron flux by extremely weak environmental EM field effects, it is interesting to note that universal enzymes such as superoxide dismutase, which protect living organisms – including us – from ROS, are phylogenetically invariant. The enzyme catalase, that decomposes H2O2 about 106 times faster than Fe2+ alone, is present in virtually all organisms, aerobic and anaerobic. This suggests that even at the time, when the early Earth environment was still pervasively reduced, early Life had to adapt to an environmental oxidative “driving force” that created an omnipresent oxidative stress. Several biochemical mechanisms exist now in living cells, and seem to have existed for a very long time, to enzymatically counteract ROS so as to safeguard the time-keeping function of the dynamic redox cycle. Specifically, they affect chemical intermediates that are inherently reactive, unstable, and self-regenerating, thereby directly controlling the oscillating induction and suppression of RNA transcription in more than 80% of all genes. Preliminary results, which we obtained during the last year, are consistent with the proposed hypothesis that the ambient EM field is an environmental unit factor for time-setting, a kind of “tuning fork” for universal biochemical reactions in living cells. We found that extremely small intensity and frequency variations in SR modulate the electron flux through the electron transport chain in living organisms. Specifically, we have developed an EM attenuation chamber with a dynamic feedback loop electronics that enable selectable, real-time cancelation of the ambient EM field. Using this chamber, we have demonstrated that attenuation of the Earth’s ambient magnetic field not only reduces the yeast cell division rate, assayed by the budding index, but also significantly reduces basal lipid oxidation likely due to reduction of basal electron leakage from the ETC. For example, using a cell buoyancy assay, we have demonstrated that yeast cells grown inside the EM attenuation chamber contain lower amounts of saturated lipid in their cell membranes than yeast cells grown in parallel ambient EM field. Additionally, to help demonstrate that extremely weak EM field can attenuate electron dwell time in vivo, we have used the electron resonance-specific frequency at magnetic flux intensities of 200 nanoTesla (about 1/2000th the intensity of the Earth’s magnetic field) to inhibit yeast cell growth by more than 50%. These results indicate that the ambient EM field couples effectively to the electron transport in vivo, thereby validating our approach to control the ambient EM field using a custom-built EM chamber with dynamic feedback loop electronics that enables selectable modulation of ambient EM field. Using our preliminary results, we have submitted a grant proposal entitled “Disruption of biorhythms in spaceflight by time varying electromagnetic radiation” in response to NASA solicitation NNH14ZTT002N: Research Opportunities for Flight Experiments in Space Biology (ILSRA). Its goal is to use space-qualified hardware on the International Space Station (ISS) to study metabolically synchronous, continuously grown yeast culture in media that contains a photonic reporter of magnetic field effects on metabolic rate in spaceflight above the ionosphere, i.e. above 400 km, where SR waves are absent. We also continue a computational study of the effects of SR field perturbations on the metabolic rate. For example, since the Earth-ionosphere cavity is distorted by the moon's gravity (as well as the sun's gravity, which affects the spring and neap tide cycle), we evaluated periodic changes in the Earth's magnetic field - continuously recorded over several months - that vary with the position and phase of the moon. Using extremely high-resolution magnetic field data from several globally stationed field magnetometers –acquired at 130 Hz sampling rate – we have developed efficient computational algorithms for time series analysis of SR field variations and their correlation with physiological variables. We plan to continue to further develop these EM data mining tools to validate EM coupling between organisms and the ambient field. We anticipate that the periodic and episodic changes in the Earth’s waveguide geometry will produce variations in the SR frequencies, which can help explain published reports on environmentally induced phenomena in animal physiology, human health and epigenetics. We have also performed extensive calculations concerning EM effects on protein molecules. Specifically, we have developed algorithms for the computational analysis of the ambient ELF and ULF coupling with the ATP Synthase rotor enzyme, which has analogous structures in existing technology. Intriguingly, ATP synthase is a molecular motor that works at nearly 100% efficiency, rotating discretely between 11 to70 Hz. Such rotational frequencies correspond to ATP hydrolysis rates of 75–150 ATP s-1. At high concentration of ATP (>10 μM), the free rotational rates are saturated at 11Hz because of viscous friction. Because the ATP synthase rotation moves positively charged protons across the membrane, and also moves charged amino acids orthogonally relative to the membrane voltage potential, the enzyme rotation produces local EM fields in the SR band. Therefore, we believe that the ATP synthase can resonate with the SR field as previously described even for individual atoms that have the capability to absorb EM energy from a spatial region that is orders of magnitudes larger than their own volume. This astonishing phenomenon is probably quantum-mechanical in nature. It is possible because the electric dipole moment induced on an atom by an incident EM field, absorbs much more energy, per unit time, than is flowing through its geometrical cross section. It is widely recognized that ATP synthase operates as a molecular motor. Structures such as the rotors and the stator arm derive their names from structural units in electric motors. Over the course of our investigation, we have identified several systems, including stepper motors, rotary solenoid, homopolar motors/generators, that could have features similar to the ATPase and could be used for modeling. We already performed numerical simulations of the torque from SR to determine whether or not they can provide the energy, or a good fraction of the energy, needed to overcome the potential barrier to ATP synthesis above the activation energy level. Our work demonstrates previously unrecognized and un-described evidence for quantum effects in biology by allowing effective EM coupling well below the thermal noise level to the universal electron transport chain (ETC) in vivo. Based on this discovery, we plan to further develop an EM chamber for the selectable modulation of the ambient SR field, based on active cancelation, to enable the study of processes to slow down life reversibly without death. The proposed “Suspended Animation Chamber” is revolutionary because no other method can actively account for the quantum effects of SR on the fundamental life-sustaining process of electron transport in vivo.
Bibliography Type:	Description: (Last Updated: 07/01/2025)

Abstracts for Journals and Proceedings	Stolc V, Freund F. "The physical basis for induction of genetic variation by oxidation." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3212.pdf , Feb-2014
Articles in Peer-reviewed Journals	Freund F, Stolc V. "Nature of pre-earthquake phenomena and their effects on living organisms." Animals. 2013;3(2):513-31. http://dx.doi.org/10.3390/ani3020513 , Jun-2013